
Part I

Aerobic Versus Strength Training



The Functional Genome in Physical Exercise

2

Wilhelm Bloch

Introduction

Different kinds of physical exercise such as aerobic, anaerobic, and resistance training as well as the combination of these trainings essentially contribute to an enhancement of physical performance in elite and recreational sports. Furthermore, physical activity leads to a risk reduction, better prognosis, and decrease of specific medical treatment side effects of several common diseases, including cancer, cardio-vascular-, metabolic-, and neurodegenerative disorders [1–3]. Recent research suggests that exercise acts as a potent regulator of the functional genome through epigenetic modifications. The functional genome may explain short and long-lasting variations in health and performance in relation to the physical activity and training. Moreover, the functional genome is a product of the genome, including polymorphisms, and epigenetic modulations of gene availability. Therefore, it is necessary to consider gene polymorphisms and epigenetic modifications as well as the combination of both in order to understand the functional genome underlying the individual performance.

Polymorphisms/Genome

The genome is derived from the blueprint of the human organism in the form of the deoxyribonucleic acid (DNA) sequence of the 23 pairs of chromosomes found in every nucleated cell and the genes encoded in the mitochondrial DNA. A chromosome is formed by two complementary strands of DNA. DNA molecules are large polypeptides in which the backbone of the molecule is composed of five-carbon sugar residues, i.e., deoxyribose. The genetic information of each chromosome is

W. Bloch
Department of Molecular and Cellular Sport Medicine, German Sport University Cologne,
Cologne, Germany
e-mail: w.bloch@dshs-koeln.de

stored in a long string of the four DNA bases: adenine (A), cytosine (C), guanine (G), and thymine (T). The order and number of the bases determine the information content of each gene, coding the blueprint for specific proteins. Every gene consists of coding sequences (exons), noncoding regions (introns), and regulatory sequences. The genes can reveal variations in the base sequence of the DNA [4]. DNA sequence variations that are common in the population are called polymorphisms, as opposed to rare gene variations that are called mutations. The less common base must have a frequency of at least 1% in the population. The most common type of genetic variations among individuals is a single nucleotide polymorphism, mostly described as SNPs (pronounced “snips”). Single bases are replaced by another base leading to a change of the genetic sequence. Such variations of the DNA sequence can affect the expression of the gene and alter the products coded by this gene. Furthermore, these gene variants may explain differences in the individual phenotype. The SNPs may reveal differences in physical capabilities and training-induced effects between subjects. Therefore, research has focused for more than 20 years on recognizing polymorphisms relevant for the prediction of physical capacity and exercise engagement [4], even though it is obvious that such polymorphisms cannot sufficiently explain the human phenotype or the exercise-related individual variants.

Epigenetics/Functional Genome

The basis for understanding the relevance of the functional genome was provided by Conrad Waddington. He defined it as “The branch of biology which studies the causal interactions between genes and their products, which bring the phenotype into being” [5]. Today, we know that the epigenetic mechanisms behind the regulation of the functional genome are “causal interactions” consisting of three major mechanisms:

1. DNA modifications by methylation of bases which do not affect the base sequence
2. Posttranslational modifications of histone proteins by different mechanisms such as acetylation, methylation, and phosphorylation
3. Expression of small RNA molecules, also known as micro (mi) RNAs.

The cytosine methylation within the DNA is the leading mechanism for longtime changes of the functional genome. Such modifications can alter the functional genome for different time periods, from short (minutes to days) to long (weeks to years). Furthermore, it becomes obvious that these changes are partially stable throughout the lifetime and also across generations. A hypermethylation in cytosine-rich regions (i.e., called CpG islands) correlates with gene suppression and modulation of the transcriptional activity of the gene, depending on the localization of the methylated gene side. Methylation of CpG islands in the promotor region makes the DNA inaccessible for transcriptional-relevant factors. Furthermore, it can lead to recruitment of enzymes, provoking further inhibitory epigenetic modifications. DNA-methyltransferases (DNMTs) are the enzymes which perform the DNA-methylation [6].

A second mechanism regulating the activity of gene availability is the histone packing of the DNA-strand. This mechanism is essential to condense the DNA-strand in a relatively small nucleus without a loss of DNA availability. Histone proteins and the adjacent DNA are called chromatin. The histone isoforms 2A, 2B, 3, and 4 build an octameric structure where the DNA is recoiled on with 2.5 turns each. The DNA part (Linker DNA) which connects two of those complexes is stabilized by histone 1 proteins. Electrostatic attraction forces of the negatively charged DNA-backbone and positively charged amino-acid side-chains in the N-terminus of histone proteins are responsible for the DNA/histone protein connection. The connection can be prevented by neutralization of the binding force through positive charges between these amino-acid side-chains. Neutralization of binding force leads to a less compact and better accessible DNA and/or produce recruiting sites for gene-activating and silencing proteins. The modifications are performed by specific enzymes such as histone acetyl transferases (HATs) and histone deacetylases (HDACs) [7, 8]. HATs and HDACs are mediating the modulation of the acetylation stage of the histones.

The third epigenetic mechanism underlying the DNA-methylation and histone modifications is the expression of miRNAs which does not regulate the DNA availability but the stability of the transcription products, i.e., the messenger mRNAs. MicroRNAs are short RNA molecules (21 bases on average) which can bind to complementary messenger (m)RNA, thereby inhibiting the translation and inducing mRNA degradation [9]. In addition to the classic role suggested for miRNAs, new research revealed that they are also capable of activating translational processes [10].

The first descriptions of epigenetic alterations were in context of imprinting and X-chromosome silencing. Epigenetic regulations were primarily understood as static and long-lasting alterations. Today, the understanding changed towards a modulation of the functional genome by epigenetic mechanisms. Depending on the type of epigenetic alterations, these were considered roughly dynamic. Especially histone modifications and expression of miRNAs are rather dynamic processes. They are highly sensitive to intrinsic and extrinsic factors leading to short up to long-lasting alterations of gene and gene product availability. It is, however, important to bear in mind that epigenetic modifications are mainly gene- and tissue-specific alterations with different time courses [11]. Therefore, it is not surprising that epigenetic modifications by different types of stimuli (i.e., different training stimuli) are mainly shown for all organs, tissues, and cells in the mammalian organism. Furthermore, changes of the functional genome by epigenetic modifications are important for physical performance and several chronic diseases which can be modulated by physical exercise [12].

Interaction of Polymorphisms and Epigenetic Modulations

Changes in the DNA sequence, e.g., by polymorphism are the major cause for gene regulations. On the other hand, chromatin structure regulates gene activity at the epigenetic level. If polymorphism results in punctual depletion of a methylable site

and this has spreading effects on adjacent sequences, one would expect to observe an association between the DNA-methylation state and proximal DNA sequence. In addition, histone-dependent epigenetic regulations are influenced by polymorphism showing the complex interaction between genome and epigenome. A further layer of complexity in the interplay between both genome and epigenome exists in the miRNA. It is indicated that polymorphisms can impact on miRNA function [13]. Therefore, it can be assumed that the functional genome can only be understood if epigenetic modulations are related to the structural genome including the polymorphisms and vice versa.

Physical Exercise Changes the Functional Genome Dependent on Polymorphisms and Epigenetic Modulations

The discussion on the relevance of genome and epigenome in physical performance may be easier and understood if the focus is set on the ACE gene. The ACE I/D polymorphism (i.e., insertion/deletion polymorphism) is related to endurance performance. However, consistent with studies in which genetic associations between the ACE gene and the corresponding I/D polymorphism were shown; it appears that epigenetic modulations also affect ACE gene activity, both with and without the ACE polymorphism. It is speculated that the epigenetic regulation of the ACE gene is as relevant to human endurance performance as the I/D polymorphism [14]. On the other hand, physical activity affects health and physical fitness/performance by epigenetic modulation of the functional genome, which may aid explaining long-lasting effects of physical activity.

About 10 years ago, Pedersen and colleagues [15] revealed that acute physical exercise leads to a short inflammatory-like cytokine pattern (particularly an increase in Interleukin (IL)-6) which is followed by mid-term anti-inflammatory response (i.e., an increase in IL-10, depression of TNF- α production, and expression of soluble receptors of pro-inflammatory cytokines). Therefore, they proposed a model which stated that regular exercise has a chronic anti-inflammatory effect, possibly explaining the preventive effects of exercise against several chronic diseases. Furthermore, numerous studies reported decreased serum levels of the inflammatory acute phase protein C-reactive protein (CRP) after different kind of exercise interventions [16]. Such mid- and long-term alterations of the inflammatory status could be related to epigenetic modifications [12]. Therefore, it is not surprising that the group of Nakajima et al. [17] showed that the ASC gene (a gene which encodes for pro-inflammatory cytokines) reveals an age-dependent loss of methylation in the promotor region. The consequence of this hypomethylation is an increased expression of pro-inflammatory cytokines in older subjects and can be partially restored by a 6-month interval endurance exercise program Nakajima et al. [17]. The higher levels of methylation in the promotor region of the ASC gene compared to sedentary controls can lead to a decreased expression of pro-inflammatory cytokines. Despite these examples, several exercise-induced alterations of the functional genome by epigenetic mechanisms are reported.

Skeletal Muscle

Epigenetic regulation, such as DNA-methylation, histone modifications, and microRNAs, are believed to be crucial to morphological changes. DNA-methylation affects the expression of many genes that are critical to skeletal muscle development, such as the homeobox genes, T-box genes, and sine oculis-related homeobox 1, which is strongly hypermethylated, whereas contractile fiber genes are hypomethylated. Furthermore, exercise induces a whole genome hypomethylation in human skeletal muscle and a dose-dependent expression of the pyruvate dehydrogenase kinase isozyme 4, of the peroxisome proliferator activator receptor delta and of the glucose transporter GLUT4, which counteracts age and diseases-dependent hypermethylation [18]. Beside several further examples for histone modifications by exercise, McGee and colleagues [19] detected HDAC-specific inhibition patterns and changes in histone acetylation after a single bout of exercise. While HDAC4 was translocated from the nucleus to the cytosol, HDAC5 showed elevated ubiquitination levels through exercise. Furthermore, high blood lactate levels induced by exercise may alter histone acetylation and affect p38 MAPK signaling, gene expression, and thereby cell differentiation and adaptation in myoblasts and skeletal muscle in vitro and likely in vivo [20]. Safdar et al. [21] showed increased levels of miRNA 1 as well as a decreased expression of its target HDAC4 in exercising mice. Since HDAC4 is known to be a transcriptional repressor for muscle-specific differentiation factors (e.g., MyoD), this study illustrates the complexity of epigenetic modifications and its own epigenetic regulation. Further studies aimed at providing a better understanding of DNA-methylation, histone modulation, and miRNA regulation in muscle development and homeostasis may help to better understand the role of physical exercise for skeletal muscle maintenance, growth, adaptation, and repair.

Vascular System

The functioning and health of the vascular system is dependent on vascular smooth muscle cell (SMC), and endothelial cell (EC) structure and function as well as on extra cellular matrix (ECM). The phenotype of the SMCs and ECs as well as the composition of ECM plays a key role in the protection against atherosclerosis and other vascular diseases. SMCs and their products contribute to almost 70% of the mass of lesions in atherosclerotic events [22, 23]. They are highly adaptive in response to environmental alterations [24]. Epigenetic modifications are involved in the adaptation of SMCs phenotypes [25]. Several examples can be given for the epigenetic regulations of the vascular system, all of which are relevant for the explanation of changes of the functional genome. For example: (1) Specific histone modifications, like the acetylation of H3K9 and deacetylation at different lysine-residues in H4 induce the binding of SRF to its DNA binding-site leading to alteration of proliferation of SMCs. (2) Myocardin (a SRF-co-activator protein) recruits different types of HATs and HDACs which may lead to the modifications mentioned above

[26, 27]. (3) Epigenetic modifications in the superoxide dismutase gene lead to SMC proliferation and a decreased apoptosis rate [28, 29]. (4) Increased oxidative stress levels are associated with an altered activity of HDACs and HATs [30]. (5) Matrix-metallo-proteinases (MMPs) which have the potential to reorganize the extra cellular matrix (ECM). Besides MMP-2 and -9 the expression of other MMPs -1, -3, -13 also underlies an epigenetic regulation [31]. In addition to exemplary described DNA- and histone modifications, miRNAs are critically involved in hypertensive-induced pathological changes of SMCs [32]. Although it is well known that exercise can influence all of the described mechanisms, less is known about the impact of exercise on epigenetic modifications of the vascular system in humans. Exercising mice revealed altered endothelial function, leading to significantly lower levels of angiotensin 2, endothelin 1, plaques, and foam cells compared to sedentary animals in a high-fat diet atherosclerosis model. These modulations were associated with changes of miRNA expression (miRNA 155 up-regulation; miRNA 146, and miRNA 126 down-regulation) in trained animals [33]. Interestingly, the expression of vasodilatory eNOS is known to be inhibited by miRNA 155 [34] providing evidence for an exercise-induced epigenetic regulation of endothelial function with regard to NO production. A down-regulation of miRNA 126 leads to endothelial dysfunction and suppresses the expression of integrins which is followed by impaired properties for leukocyte extravasation [35]. Therefore, the exercise-triggered increase in miRNA 126 might also be a positive epigenetic effect. It seems interesting for further studies to relate exercise-mediated shear stress with epigenetic regulation of eNOS as well as further epigenetic gene modulations by exercise stimuli in endothelial cells [36]. Furthermore, exercising mice revealed decreased levels of miRNA 16 and 126 which are both known to suppress the expression of VEGF [37] and thereby vascular growth, repair, and remodeling. Regarding endothelial and vascular repair, recent studies demonstrated that epigenetic mechanisms are important transcriptional regulators of angiogenic genes in endothelial cells (including modifications of DNA and histones as well as noncoding RNA) [12]. Additionally, they play an important role in the regulation of endothelial stem/progenitor cell function [38] but up to now, a direct link between epigenetic modulations of endothelial cells and endothelial stem/progenitor cells is missing.

Heart

The functional genome plays an important role in the physiological and pathophysiological development and adaption of cardiomyocytes in the context of adaptive and maladaptive cardiac-hypertrophy and function/dysfunction. Epigenetic mechanisms seem crucial for the determination of the functional genome in cardiomyocytes. Animal studies indicate that the inhibition of HDAC3 leads to the development of cardiomyopathies in a time point and nutrition-dependent manner [39]. The cardiomyopathy may develop more severe, if HDAC are early inhibited and a high-fat diet is given, ultimately leading to increased lethality. Furthermore, the inhibition of

HDAC4 promotes cardiac-stem-cell-induced regeneration and repair [40]. A miRNA-mediated induction of cardiac-hypertrophy is also reported in numerous studies [41]. On the chromatin level, human studies have not investigated the impact of exercise on cardiomyocytes, while first evidence for epigenetic regulation of chromatin is recently given in mice. It has been demonstrated that exercise can alter the HDAC4 level influencing the occurrence of heart failure and cardiac fatigue [42]. With regard to miRNAs, it has been reported that exercise decreased cardiac miRNA 208a expression in mice [43]. Further speculation for an exercise-dependent regulation of cardiomyocytes by miRNA is given for miRNA208a, which leads to a reversion of pathological changes in myosin heavy chain (MHC) expression [44]. The pathological changes are characterized by an overexpression of the fetal β MHC, leading to slow ATPase activity as well as a down-regulation of the adult α MHC (fast ATPase activity). The fact that exercise improves the balance of α MHC and β MHC was already described earlier but the results of these studies reveal that the underlying mechanism is potentially based on epigenetic alterations.

Extracellular Matrix, Adipocytes, and Immune Cells

Beside maladaptation of cardiomyocytes, pathophysiological alterations of the ECM are a hallmark of several diseases including cardiac diseases. Changes of the ECM are mostly characterized by an accumulation of collagen, induced by either an overexpression or decrease in collagen degradation through MMPs and an altered function of fibroblasts (e.g., cytokine production, such as TNF- α and TGF- β). Furthermore, changes of DNA-methylation are related to these mechanisms [45] and numerous miRNAs were also described to be involved in these fibrotic processes. Exercise is known to directly impact the ECM in different tissues and may alter the collagen pattern (e.g., in the heart of exercising rats by down-regulation of several active MMPs), while it seems to counteract an age-dependent increase in TGF- β [46]. Furthermore, exercise induces an up-regulation of miRNA 29c, which has been recognized to be involved in fibrotic alterations. Since this increased miRNA 29 c expression was accompanied by a decreased expression of collagen I and III, further evidence for a positive exercise-induced epigenetic regulation is provided [47].

Besides fibroblasts, other mesenchymal derived cell types, such as adipocytes are also epigenetically regulated. In adipocytes a 6-month exercise program can induce an increased hypermethylation of HDAC4 and a reduced expression of HDAC4. This could counteract obesity by reduction of lipogenesis in adipocytes [48]. Furthermore, other bone marrow-derived non-mesenchymal cell types are targets for exercise-induced epigenetic regulation. It has been suggested that a 4-week high intensity exercise program in young healthy adults has a sensitive impact on the methylome of leukocytes. In addition, the demethylation of numerous CpG islands by exercise leads to an epigenetic activation of lymphocytes [49]. Actually, the same group also showed that resistance exercise training improves muscular strength is associated with reprogramming of the leukocyte DNA methylome and transcriptome [50]. These results support the findings from other studies, which

described an exercise-induced epigenetic regulation of leukocytes [17, 51, 52]. These findings reveal a distinct influence of exercise for the functional genome on bone marrow-derived mesenchymal and hematopoietic cell lines.

Summary

As mentioned above, both the genome and epigenome are interacting to determine the individual human phenotype including characteristics of physical fitness, performance, and health. However, the knowledge about this interplay and the underlying mechanisms are still sparse. In particular, the knowledge about the regulation of the functional genome by the type of physical exercise is rare. Most epigenetic studies do not differentiate between types of physical exercise, although it is well known that aerobic and strength training induces different mechanisms and signaling, potentially relevant for the determination of the functional genome. Furthermore also the predictive value of the functional genome including polymorphism and epigenetics for aerobic and strength performance and trainability is not yet sufficiently deciphered. Nevertheless, distinct evidence suggests that besides epigenetic drugs, physical activity has the potential to regulate the human phenotype by altering the functional genome as well as the product of the genome and epigenome. As such it becomes evident that the functional genome builds the foundation for biological adaptations to exercise training. Sophisticated research strategies are needed to transfer study results from bench to practice. These strategies must respect the type and mode of exercise, as well as the underlying mechanisms and signaling.

Acknowledgments The author would like to thank Mrs. Christine Koliamitra for editorial support.

References

1. Lee IM, Shiroma EJ, Lobelo F, Puska P, Blair SN, Katzmarzyk PT, Lancet Physical Activity Series Working Group. Effect of physical inactivity on major non-communicable diseases worldwide: an analysis of burden of disease and life expectancy. *Lancet*. 2012;380(9838):219–29.
2. Mattson MP. Lifelong brain health is a lifelong challenge: from evolutionary principles to empirical evidence. *Ageing Res Rev*. 2015;20:37–45.
3. Schmid D, Leitzmann MF. Association between physical activity and mortality among breast cancer and colorectal cancer survivors: a systematic review and meta-analysis. *Ann Oncol*. 2014;25(7):1293–311.
4. Bouchard C, Rankinen T, Timmons JA. Genomics and genetics in the biology of adaptation to exercise. *Compr Physiol*. 2011;1(3):1603–48.
5. Waddington CH. The epigenotype. *Endeavour*. 1942;1:18–20.7.
6. Klose RJ, Bird AP. Genomic DNA methylation: the mark and its mediators. *Trends Biochem Sci*. 2006;31(2):89–97.
7. Kouzarides T. Chromatin modifications and their function. *Cell*. 2007;128:693–705.
8. Kouzarides T, Berger SL. Chromatin modifications and their mechanism of action. In: Allis CD, Jenuwein T, Reinberg D, editors. *Epigenetics*. 1st ed. New York: Cold Spring Harbor Laboratory; 2006. p. 191–209.

9. Mercer TR, Dinger ME, Mattick JS. Long non-coding RNAs: insights into functions. *Nat Rev Genet.* 2009;10(3):155–9.
10. Lee S, Vasudevan S. Post-transcriptional stimulation of gene expression by microRNAs. *Adv Exp Med Biol.* 2013;768:97–126.
11. Jenuwein T, Allis CD. Translating the histone code. *Science.* 2001;293:1074–80.
12. Zimmer P, Bloch W. Physical exercise and epigenetic adaptations of the cardiovascular system. *Herz.* 2015;40(3):353–60.
13. Zaina S, Pérez-Luque EL, Lund G. Genetics talks to epigenetics? The interplay between sequence variants and chromatin structure. *Curr Genomics.* 2010;11(5):359–67.
14. Raleigh SM. Epigenetic regulation of the ACE gene might be more relevant to endurance physiology than I/D polymorphism. *J Appl Physiol* 2012;112:1082–3.
15. Pedersen BK, Febbraio MA. Muscle as an endocrine organ: focus on muscle-derived interleukin-6. *Physiol Rev.* 2008;88(4):1379–406.
16. Hayashino Y, Jackson JL, Hirata T, Fukumori N, et al. Effects of exercise on C-reactive protein, inflammatory cytokine and adipokine in patients with type 2 diabetes: a meta-analysis of randomized controlled trials. *Metabolism.* 2014;63(3):431–40.
17. Nakajima K, Takeoka M, Mori M, Hashimoto S, et al. Exercise effects on methylation of ASC gene. *Int J Sports Med.* 2010;31:671–5.
18. Moresi V, Marroncelli N, Coletti D, Adamo S. Regulation of skeletal muscle development and homeostasis by gene imprinting, histone acetylation and microRNA. *Biochim Biophys Acta.* 2015;1849(3):309–16.
19. McGee SL, Fairlie E, Garnham AP, Hargreaves M. Exercise-induced histone modifications in human skeletal muscle. *J Physiol.* 2009;587(Pt 24):5951–8.
20. Willkomm L, Gehlert S, Jacko D, Schiffer T, Bloch W. p38 MAPK activation and H3K4 trimethylation is decreased by lactate in vitro and high intensity resistance training in human skeletal muscle. *PLoS One.* 2017;12(5):e0176609.
21. Safdar A, Abadi A, Akhtar M, Hettinga BP, et al. miRNA in the regulation of skeletal muscle adaptation to acute endurance exercise in C57Bl/6J male mice. *PLoS One.* 2009;4:e5610.
22. Owens GK. Regulation of differentiation of vascular smooth muscle cells. *Physiol Rev.* 1995;75(3):487–517.
23. Ross R. The pathogenesis of atherosclerosis: a perspective for the 1990s. *Nature.* 1993;362:801–9.
24. Hoofnagle MH, Wamhoff BR, Owens GK. Lost in transdifferentiation. *J Clin Invest.* 2004;113(9):1249–51.
25. Alexander MR, Owens GK. Epigenetic control of smooth muscle cell differentiation and phenotypic switching in vascular development and disease. *Annu Rev Physiol.* 2012;74:13–40.
26. McDonald OG, Owens GK. Programming smooth muscle plasticity with chromatin dynamics. *Circ Res.* 2007;100:1428–41.
27. McDonald OG, Wamhoff BR, Hoofnagle MH, Owens GK. Control of SRF binding to CArG box chromatin regulates smooth muscle gene expression in vivo. *J Clin Invest.* 2006;116:36–48.
28. Archer SL, Marsboom G, Kim GH, et al. Epigenetic attenuation of mitochondrial superoxide dismutase 2 in pulmonary arterial hypertension: a base for excessive cell proliferation and a new therapeutic target. *Circulation.* 2010;121:2661–71.
29. Kim GH, Ryan JJ, Archer SL. The role of redox signaling in epigenetics and cardiovascular disease. *Antioxid Redox Signal.* 2013;18(15):1920–36.
30. Ito K, Hanazawa T, Tomita K, Barnes PJ, Adcock IM. Oxidative stress reduces histone deacetylase 2 activity and enhances IL-8 gene expression: role of tyrosine nitration. *Biochem Biophys Res Commun.* 2004;315:240–5.
31. Chen KC, Wang YS, Hu CY, Chang WC, et al. OxLDL up-regulates microRNA-29b, leading to epigenetic modifications of MMP-2/MMP-9 genes: a novel mechanism for cardiovascular diseases. *FASEB J.* 2011;25(5):1718–28.
32. Bártkai S, Thum T. MicroRNAs in hypertension: mechanisms and therapeutic targets. *Curr Hypertens Rep.* 2012;14:79–87.
33. Wu XD, Zeng K, Liu WL, Gao YG, et al. Effect of aerobic exercise on miRNA-TLR4 signaling in atherosclerosis. *Int J Sports Med.* 2014;35(4):344–50.

34. Zhang J, Zhao F, Yu X, Lu X, Zheng G. MicroRNA-155 modulates the proliferation of vascular smooth muscle cells by targeting endothelial nitric oxide synthase. *Int J Mol Med*. 2015;35(6):1708–14.
35. Zhuang Y, Peng H, Mastej V, Chen W. MicroRNA regulation of endothelial junction proteins and clinical consequence. *Mediat Inflamm*. 2016;2016:5078627. <https://doi.org/10.1155/2016/5078627>.
36. Gielen S, Sandri M, Erbs S, Adams V. Exercise-induced modulation of endothelial nitric oxide production. *Curr Pharm Biotechnol*. 2011;12(9):1375–84.
37. Fernandes T, Magalhães FC, Roque FR, Phillips MI, et al. Exercise training prevents the microvascular rarefaction in hypertension balancing angiogenic and apoptotic factors: role of microRNAs-16, -21, and -126. *Hypertension*. 2012;59:513–20.
38. Shiva Shankar TV, Willems L. Epigenetic modulators mitigate angiogenesis through a complex transcriptomic network. *Vascul Pharmacol*. 2014;60(2):57–66.
39. Montgomery RL, Hullinger TG, Semus HM, Dickinson BA, Seto AG, et al. Therapeutic inhibition of miR-208a improves cardiac function and survival during heart failure. *Circulation*. 2011;124:1537–47.
40. Zhang LX, DeNicola M, Qin X, Du J, et al. Specific inhibition of HDAC4 in cardiac progenitor cells enhances myocardial repairs. *Am J Physiol Cell Physiol*. 2014;307(4):C358–72.
41. Nguyen MA, Karunakaran D, Rayner KJ. Unlocking the door to new therapies in cardiovascular disease: microRNAs hold the key. *Curr Cardiol Rep*. 2014;16(11):539.
42. Lehmann LH, Jebessa ZH, Kreusser MM, Horsch A, et al. A proteolytic fragment of histone deacetylase 4 protects the heart from failure by regulating the hexosamine biosynthetic pathway. *Nat Med*. 2018;24(1):62–72.
43. Soci UP, Fernandes T, Rosa KT, Irigoyen MC, et al. The role of microRNA-208a in cardiac hypertrophy. *Clin Sci*. 2016. pii:CS20160480 Epub ahead of print.
44. Montgomery RL, Potthoff MJ, Haberland M, et al. Maintenance of cardiac energy metabolism by histone deacetylase 3 in mice. *J Clin Invest*. 2011;118(11):3588–97.
45. Tao H, Yang JJ, Shi KH, Deng ZY, Li J. DNA methylation in cardiac fibrosis: new advances and perspectives. *Toxicology*. 2014;323:125–9.
46. Kwak HB. Aging, exercise, and extracellular matrix in the heart. *J Exerc Rehabil*. 2013;9(3):338–47.
47. Soci UP, Fernandes T, Hashimoto NY, Mota GF, et al. MicroRNAs 29 are involved in the improvement of ventricular compliance promoted by aerobic exercise training in rat. *Physiol Genomics*. 2011;43:665–73.
48. Rönn T, Volkov P, Davegårdh C, Dayeh T, et al. A six months exercise intervention influences the genome-wide DNA methylation pattern in human adipose tissue. *PLoS Genet*. 2013;9(6):e1003572.
49. Denham J, O'Brien BJ, Marques FZ, Charchar FJ. Changes in the leukocyte methylome and its effect on cardiovascular related genes after exercise. *J Appl Physiol* (1985). 2015;118(4):475–88.
50. Denham J, Marques FZ, Bruns EL, O'Brien BJ, Charchar FJ. Epigenetic changes in leukocytes after 8 weeks of resistance exercise training. *Eur J Appl Physiol*. 2016;116(6):1245–53.
51. Zimmer P, Bloch W, Schenk A, Zopf EM, et al. Exercise-induced natural killer cell activation is driven by epigenetic modifications. *Int J Sports Med*. 2015;36(6):510–5.
52. Zimmer P, Baumann FT, Bloch W, Schenk A, et al. Impact of exercise on pro inflammatory cytokine levels and epigenetic modulations of tumor-competitive lymphocytes in Non-Hodgkin-Lymphoma patients-randomized controlled trial. *Eur J Haematol*. 2014;93(6):527–32.